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RESEARCH ARTICLE

Spatial versus normal audio guides in exhibitions: Cognitive mechanisms and effects on learning

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Abstract
The present paper examines possible benefits of spatial audio guides on learning outcomes in the spatial learning setting of an experimental exhibition and attempts to differentiate between different mechanisms underlying such an effect. In Experiment 1, we examined whether the spatial contiguity principle may be such a mechanism. A spatial audio guide was used that allows for perception of audio explanations either at the location of the exhibit (audio-visuo-spatially close) or within the headphones (audio-visuo-spatially distant). The results showed that with the external, exhibit-located compared to headphone-located presentation, memory of audio explanations was significantly better. No significant differences were found regarding text-picture-integration. However, these results cannot only be explained by audio-visuo-spatial contiguity but also by externalization and audio-spatial cueing. In Experiment 2, we therefore tried to disentangle these possible mechanisms and compared three conditions (audio explanations perceptually located within the exhibits, behind the visitors at the opposite wall, always at the same location in the room), thereby keeping externalization constant while varying audio-visuo-spatial contiguity and audio-spatial cueing. No significant differences between the conditions were found regarding memory of the audio explanations and text-picture integration, whereby memory of audio explanations in all conditions was at the same level as in the picture-located audio condition in the first experiment. This excludes spatial contiguity effects and audio-spatial cueing as explanations for the results in Experiment 1 and indicates a general externalization effect needing closer examination. The results are discussed regarding future studies and the use of spatial audio guides in museums and exhibitions.

Keywords Spatial audio guides · Audio-visuo-spatial contiguity · Externalization · Audio-spatial cueing · Multimedia · Exhibition

Immersive technologies such as virtual reality (VR) and augmented reality (AR) have found their way into informal learning settings, such as museums and exhibitions (Zhou et al., 2022). While they offer visitors informative and entertaining experiences, attention has to be paid not to overwhelm the visitors and distract them from the exhibits (Aitamurto et al., 2018). A promising technique in this regard is audio augmented reality (AAR), which adds a spatial dimension to auditorily presented content (Kaghat et al., 2020; Tsepapadakis

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et al., 2022). With the help of AAR, audio elements such as speech, music, sounds, and sound effects are integrated into the spatial arrangements of galleries (Giariskanis et al., 2022; Yang & Sörös, 2019), contributing to the visitors' experience of spatial and social presence (Larsson et al., 2010).

With spatial, unlike normal, audio guides, for example, audio explanations presented via headphones can be assigned a specific spatial position in the gallery giving the impression that the sound sources are located at the respective positions outside the head (e.g., emanating from a particular exhibit; Kaghat et al., 2009; Yang & Sörös, 2019), rather than inside the head—a phenomenon called sound externalization (Best et al., 2020; Plenge, 1974). Spatial audio guides also allow for audio explanations to be played automatically as soon as a certain area in the exhibition room is entered (Yang & Sörös, 2019; Zimmermann & Lorenz, 2008). Thus, visitors become active recipients of the audio explanations because both intensity and extent of the acoustically presented information can be influenced by head movements, approaching, and turning away from the spatial position where the virtual sound source is located (Zimmermann & Lorenz, 2008).

While previous research on the use of spatial audio guides in museums and exhibitions has mainly focused on the technical implementation, usability, and the potential immersive effects (D'Auria et al., 2015; Heller & Borchers, 2015; Kaghat et al., 2009, 2020; Larsson et al., 2010; Tsepapadakis et al., 2022; Zimmermann & Lorenz, 2008), the present two studies focus on the influence of spatial audio guides on cognitive processes relevant for learning, thereby especially referring to the spatial contiguity principle deduced from the cognitive theory of multimedia learning (Mayer & Fiorella, 2021).

Spatial audio perception and learning

Regarding the perceptual processing of single and simple sound patterns as well as the perception of different simultaneous sounds, Bregman (1990) proposes a perceptual process called the auditory scene analysis (ASA), describing auditory perception and how the human auditory system organizes sound into meaningful and separable elements. ASA is based on the principle that the sound of objects consists of signal-inherent components, such as its frequency and timing, which overlap with other sound sources. The auditory system must decide which signal components belong to which source and then separate this as a so-called stream, thereby enabling the localization and identification of the sound sources. This is done by grouping mechanisms based on Gestalt principles (Sutojo et al., 2020). For example, frequencies that match harmonically or change simultaneously in loudness or location are perceived as belonging to one source. The approaches of ASA are highly relevant to the perception of sounds, speech, and even room acoustic scenes.

In addition to ASA, the association model by Theile (1980) describes the cognitive processing of sounds in space and assumes a peripheral stage of the audio sensory input and a subsequent hierarchical perception consisting of a location-determining pattern selection and a higher-level, gestalt-determining pattern selection that leads to the cognitive auditory event. This dichotomy represents the two main domains of an auditory experience: place and signal—both always occurring together. Theile (1980) further proposes an interacting nonspecific system that accomplishes guidance of association and is related to short-term and long-term memory. Both short-term and long-term auditory memory are intimately associated with sound localization and the perception of sounds being located outside the head (sound externalization, Plenge, 1972). The long-term memory gradually adapts,

archiving listener-specific, invariant localization cues like spatial maps of binaural cues and spectral shapes influenced by the head and ear anatomy. Conversely, the short-term memory swiftly adjusts to context-specific auditory and environmental cues, including visual inputs and spatial dynamics. Externalization predominantly depends on the congruence of present cues with the listener's anticipatory framework stored in short-term memory, resorting to long-term memory only when contextual cues are lacking (Best et al., 2020). In spatial audio perception research, this associative steering approach is considered to be an essential theory.

While these theories mainly deal with the perception of audio information, they do not consider the aspect of audio-visual integration, meaning that the perception of audio information is influenced by visual stimuli and vice versa. According to the redundant signals effect shown in experiments with nonverbal auditive and abstract visual stimuli by Miller (1982), detection responses to audio-visual stimuli are faster than for auditive stimuli alone. This indicates that signals in two different perceptual channels contribute to a common activation that initiates responses. Similar effects could be found for speech perception. McGurk and MacDonald (1976), for example, showed that the perception of speech could be fostered by the congruent perception of lip movements. Regarding long-term memory for dynamic human interaction scenes, a study by Meyerhoff and Huff (2016) showed that audio-visual clips elicited more accurate memory performance than auditory clips. Furthermore, this advantage of audio-visual integration increased with semantic audio-visual congruency. Memory performance remained intact when auditory and visual information was presented asynchronously but decreased when the matching audio and visual tracks were presented separately with intervening tracks in between. Hence, in terms of long-term memory performance, audio-visual integration was sensitive to semantic congruency but robust to asymmetries between different modalities.

Regarding complex learning situations, psychological multimedia research provides several theoretical models explaining cognitive processing of audio-visual information and learning. They describe more precisely than the theories in the field of audio perception the audio-visual integration of information in working memory and its storage in and retention from long-term memory. The cognitive theory of multimedia learning (CTML, Mayer & Fiorella, 2021), for example, postulates that written and spoken words as well as pictures of a multimedia presentation are selected by attentional processes and further processed in two separate channels of the working memory—a verbal and a pictorial channel, both with limited capacities—leading to a verbal and a pictorial mental representation that is merged into an integrated mental model by the help of prior knowledge and transferred to long-term memory. While the CTML (Mayer & Fiorella, 2021) neglects the cognitive processing of non-verbal sounds, the cognitive-affective model of learning with media (CATLM, Moreno & Mayer, 2007) and the integrated cognitive affective model of learning with multimedia (ICALM, Plass & Kaplan, 2016) differentiate between non-verbal sounds and verbal narrations as auditive input. However, they still stick to the dichotomous modes of representation – pictorial and verbal—and do not consider non-verbal sound information as cognitive representations relevant for the learning process.

We therefore combined the above described theories from audio perception (Bregman, 1990; Theile, 1980) and psychological multimedia research (CTML, Mayer & Fiorella, 2021; CATLM, Moreno & Mayer, 2007; ICALM, Plass & Kaplan, 2016) into a theoretical model that considers not only the generation of a visual mental representation based on images and a verbal mental representation based on phonological and lexical codes in working memory, but also a non-verbal auditory mental representation based on acoustic and spatial codes: the model of audio-visual cognitive processing (MAVCP, see Fig. 1).

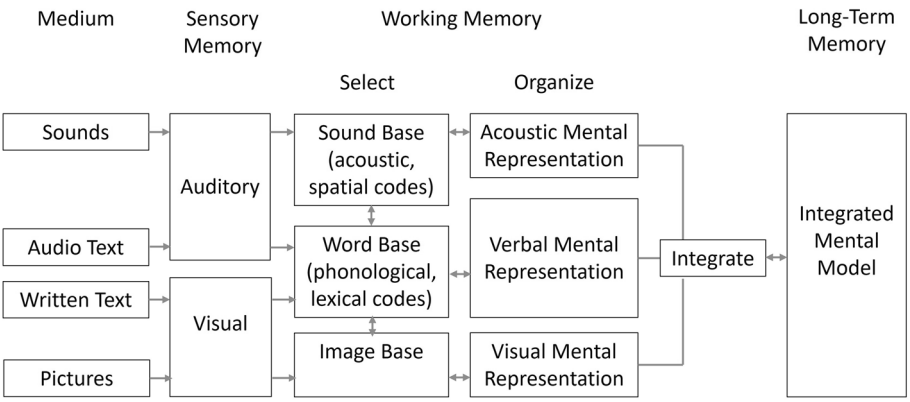


Fig. 1 The Model of Audio-Visual Cognitive Processing (MAVCP)

The MAVCP describes the cognitive processing of multimedia presentations, including non-verbal sounds, audio text, written text, and pictures. According to this model, non-verbal sounds and audio texts enter the cognitive system via one's ears and are shortly stored in auditory sensory memory, while written text and pictures enter the cognitive system via one's eyes and are shortly stored in visual sensory memory. By attentional processes, acoustic and spatial codes are selected from the auditory sensory memory and build the sound base, which is organized into an acoustic mental representation in working memory. This corresponds to the auditory scene analysis described in the field of auditory perception, in which sounds are localized and identified—the process called auditory stream segregation (Bregman, 1990; Theile, 1980). Similarly, from audio-textual information in the auditory sensory memory and from written text information in the visual sensory memory, phonological and lexical codes are attentionally selected and build the word base, which is organized into a verbal mental representation. Images are attentionally selected from the visual sensory memory and build the image base, which is organized into a visual mental representation. On the level of selection, the contents of the sound base, the word base, and the visual base are converted into each other. This means, for example, that a long deep purring stored in the sound base can be converted into the noun, cat, or the image of a cat, even when there is no cat written or visible anywhere in the environment. Finally, the acoustic, verbal, and visual mental representations are integrated into a single mental model of the multimedia presentation by the help of prior knowledge, which is then stored in long-term memory.

What is important in the MAVCP is that the quality of the resulting integrated mental model and therefore the quality of the learning outcomes depends on how clearly auditive and visual learning content is presented, how well it is extractable from the surrounding information by attentional processes, and how easily they are attributable to each other. Therefore, perceptual aspects such as audio and visual quality, auditory and visual attention, audio-visual integration, as well as the accordance with expectations play an important role for cognitive processing and learning. These should be optimized, especially in perceptually complex multimedia learning settings, such as exhibitions. Furthermore, this model emphasizes the integration not only of audio-textual (word base) and visual information (image base) but also of acoustic spatial codes (sound base) as a relevant aspect of multimedia learning. From the CTML (Mayer & Fiorella, 2021), which mainly refers to two-dimensional learning material,

different design principles have been deduced that postulate how the learning material should be designed to foster these integration processes and therefore learning. One of them is the spatial contiguity principle. Since our MAVCP is a further development of the CTML (Mayer & Fiorella, 2021) regarding spatial acoustic information, the spatial contiguity principle may also be valid for spatial learning settings including spatial acoustic information. Accordingly, then, the integration process should be easier with higher compared to lower spatial proximity between the spatial acoustic, audio-textual, and visual information. This can be implemented in complex spatial multimedia learning scenarios, such as exhibitions, by the use of the AAR technique of spatial audio guides.

The spatial contiguity principle

Regarding the integration of visual, two-dimensional learning material—written-textual (word base) and pictorial (image base) information—the spatial contiguity principle of multimedia learning, deduced from the CTML (Mayer & Fiorella, 2021) assumes that learners have better learning outcomes when they learn with learning material in which related pictorial and written textual information is presented in close proximity (Fiorella & Mayer, 2021). Using extraneous overload scenarios in which written text was either presented closely or spatially distant to the related picture detail, this effect was empirically confirmed for a broad range of learning content, such as engineering, plant biology, first aid, anatomy, or mathematics (Ginns, 2006; Schroeder & Cenkci, 2018). Meta-analyses report medium mean effect sizes, namely $d = .72$ (Ginns, 2006) and $g = .63$ (Schroeder & Cenkci, 2018). Mayer and Fiorella (2014) report a large median effect size of Cohen's $d = 1.10$. Mayer and Fiorella (2014) defined the spatial contiguity principle as a subset of the split attention principle (Ayres & Sweller, 2021). This means that learners need to invest more cognitive resources in spatially distant text and picture combinations because they have to split their attention and keep written textual information in working memory while searching for the related pictorial information and vice versa (Moreno & Mayer, 1999). In addition, effortful visual search and memory processes (e.g., Bauhoff et al., 2012; Holsanova et al., 2009; Johnson & Mayer, 2012) as well as parafoveal compared to peripheral vision (Yang et al., 2023) have been empirically confirmed to be an explanation for the spatial contiguity effect. Spatial contiguity further showed up to be most effective for learners with low prior knowledge (Mayer et al., 1995).

Nevertheless, boundary conditions may lead to varying results. Cammeraat et al. (2020) and Pouw et al. (2019) found no effect of spatial distance between texts and corresponding pictures on mental effort and learning outcomes. Based on these observations, Cammeraat et al. (2020) questioned whether the spatial contiguity effect is as robust as assumed. For this reason, it is of importance to conduct further research on the conditions of the spatial contiguity effect—in particular, dimensionality (spatial instead of two-dimensional learning settings) and modality of the learning material (audio-visual instead of only visual learning material).

Spatial contiguity in spatial settings

Up to now, only few authors have discussed the application of the spatial contiguity principle to the design of learning content in three-dimensional informal learning spaces, reporting mixed results. Myrczik (2014) argues that the design principles of multimedia learning, including also the spatial contiguity principle, should be

considered by exhibition designers in order to reduce visitors' extraneous cognitive load. Jee and Anggoro (2021) postulate that in museums and exhibitions, spatial contiguity between related items can especially foster relational thinking and learning. Grüninger et al. (2014) varied the spatial distance of conflicting texts in museums and could show that close texts compared to distant ones had a positive effect on processing depth in science but not cultural history museums. Sommerauer and Müller (2014) found that across all multimedia design principles implemented via AR technology, visitors showed a higher knowledge gain with augmented than with non-augmented exhibits. However, in a botanical garden-like learning environment, Liu et al. (2012) did not find significant differences in comprehension and learning efficiency between a condition with text and pictures on a mobile device (integrated) and a condition with text on a mobile device and real botanic objects (separated). They argued that positive motivational effects of experiencing real objects may have outweighed the expected negative effect in the spatially distant condition. Similarly, in a simulated AR experience using videos recorded in first person perspective showing plants in a botanical garden, Krüger and Bodemer (2022) did not find differences in memory performance between an integrated condition in which written information was presented as an overlay directly in front of the respective plants and a separated condition in which the written information was presented on a tablet that the filming person was holding in front of the respective plants.

However, all these studies examined the spatial contiguity principle with visual learning material. To examine whether the spatial contiguity principle is not only valid for visual learning material and the CTML but also for audio-visual learning material and the MAVCP, two experiments were conducted.

Experiment 1

Experiment 1 was preregistered and examined audio-visuo-spatial contiguity in the spatial multimedia learning setting of an exhibition showing portraits of inventors and their inventions and presenting corresponding audio explanations by a spatial audio guide. This allowed the visitors to process visual and auditive information simultaneously. Audio explanations were presented either perceptually located within the headphones, spatially distant to the portraits they referred to, or perceptually located within the portraits referred to—as if the inventors themselves were telling the visitors their story—thereby varying spatial contiguity. It was assumed that:

H1 Memory for audio explanations accompanying the portraits should be better when the explanations are perceptually located within the portraits (spatially close) than within the headphones (spatially distant).

H2 Text-picture integration should be better when the audio explanations accompanying the portraits are perceptually located within the portraits (spatially close) than within the headphones (spatially distant).

Interest in further information about the inventors and their inventions, perceived audio quality, as well as judgement of learning were assessed exploratorily.

Method

Participants

Participants were recruited from a mailing list of the first author's institute. They had to be native German speakers or have an equal level of fluency as well as normal hearing and normal or corrected vision. Participants who were studying, had studied, or exercised professions in the fields of engineering, medical engineering, mechatronics, computer science, medical informatics, or physics were excluded to keep prior knowledge low (cf. Mayer et al., 1995). Also, persons who had participated in earlier experiments with the same material and persons with high prior knowledge about the inventors and their inventions were excluded. Based on these criteria, five of $n = 59$ recruited participants had to be excluded. Further exclusions concerned two participants that needed more than three trials to pass the hearing training (description see below) and four participants with technical problems during the experimental session. The remaining $n = 48$ participants included 42 female and six male participants. They were aged between 19 and 55 years ($M = 25.85$, $SD = 7.28$) and from a wide range of disciplines (e.g., psychology, medicine, rhetoric, media science, history, politics). Based on the reported effect sizes of the meta-analyses of the spatial contiguity effect (Cohen's $d = .72$, Ginns, 2006; Hattie's $g = .63$, Schroeder & Cenkci, 2018) as well as the studies by Mayer and Fiorella (2014, Cohen's $d = 1.10$) an a priori power analysis with G*Power revealed a total of $n = 52$ participants in order to find a large effect ($d = .80$) with a power of .80 and an alpha error probability of .05. Therefore, the number of participants for the analyses was about as high as intended. A positive ethics vote of the local ethics committee was available for the study.

Design

We used a one-factorial design with spatial contiguity (audio and visual information presented spatially close vs. spatially distant) as the between-subjects factor. The participants were randomly assigned to the two different conditions.

Material

Participants visited a small experimental exhibition with six black and white photographic portraits of inventors holding their invention. The portrayed inventors were Eugene Houdry with his exhaust catalyst, Eduard Justi with his continuous current electromagnet (see Fig. 2), Wilson Greatbatch with his pacemaker, Anton van Zanten with his electronic stability program, Douglas Engelbart with his computer mouse, and Arthur Schawlow with his laser. This scientific content was chosen instead of art content because we wanted to stay close to the previous research on the spatial contiguity principle which, to our knowledge, until now, has been mainly examined with scientific learning content. Portraits of inventors and their inventions instead of just pictures of the inventions were used to motivate spatial audio. A speech directed at the participants, coming from the pictures as if the exhibits themselves were speaking to the participants, was assumed to be more plausible with the inventors speaking about their inventions compared to the inventions speaking themselves.

The portraits hung on two opposite sides (three on the left and three on the right) of the approximately 6 m × 6 m room. The order of the portraits was randomly chosen because

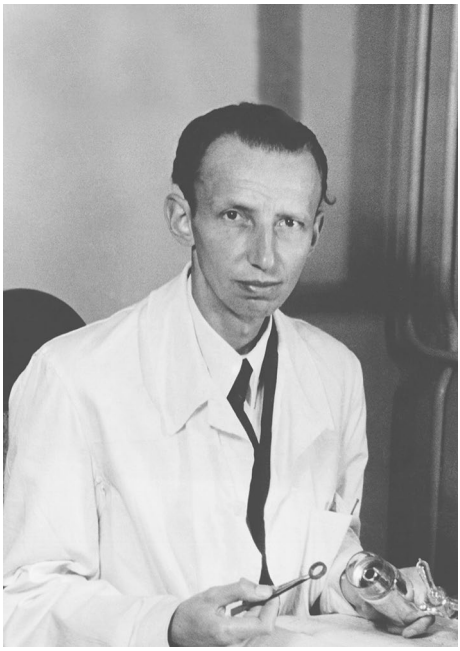


Fig. 2 Portrait of Eduard Justi holding his continuous current electromagnet. Photo: Deutsches Museum in Munich, Germany

the depicted content of each portrait was independently understandable, and, hence, a particular predefined order was not necessary. For navigation purposes, white markings were taped on the floor indicating audio areas approximately 2 × 2 m large and positioned 80 cm in front of each portrait. In each area, the corresponding audio explanation was automatically played via headphones. Participants had to visit the areas in a predefined order marked on the floor to standardize the procedure. In addition, an area in the middle of the room was marked in which the hearing training was automatically played as well as an area near the experimenter's desk in which an audio file was played for technical control purposes (see Fig. 3).

For each portrait, an audio explanation was created which was formulated in first person perspective. Each audio explanation first provided information about the professional background of the inventor, then described his invention and its functioning, and finally described its social contribution. As an example, the audio explanation about Eduard Justi and his continuous current electromagnet is provided in the appendix. The duration of the audio explanations ranged from 02:42 to 03:24 min.

The audio guide consisted of regular headphones (Sennheiser HD 600) with an HTC VIVE tracker (2018) mounted on top that were worn by the participants. The position and the direction of the participants were tracked by four HTC VIVE PRO SteamVR Base Stations 2.0 mounted at a height of approximately 2.70 m in each corner of the room. According to the position of the participant in the marked audio areas, corresponding audio explanations were automatically played via the headphones. The audio guide could be run in two modes: normal audio and spatial audio. With normal audio, the participants perceived the audio explanations to be coming from the headphones as known from using any regular stereo headphones. With spatial audio, the participants perceived the audio explanations to come out of the portraits. The two types of audio settings corresponded to the two conditions in our study.

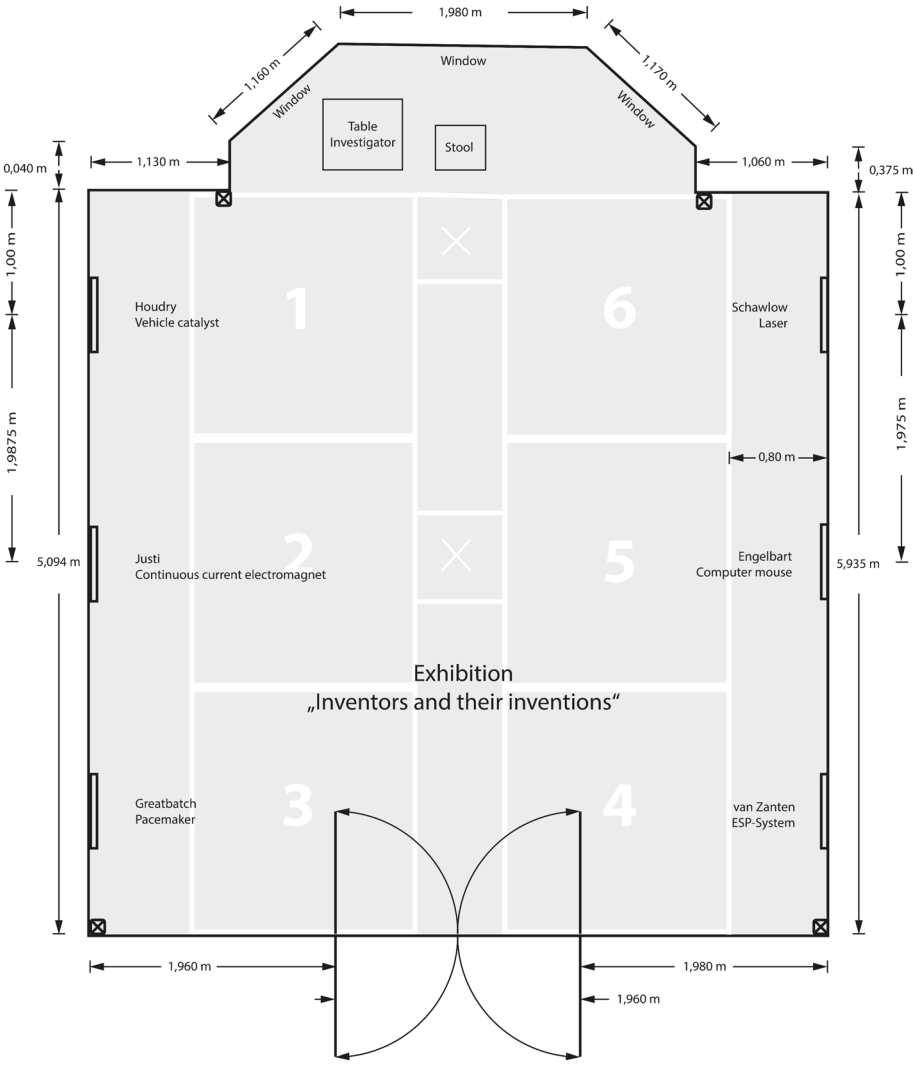


Fig. 3 Floor plan of the experimental exhibition with portraits on the left and right, as well as with white navigation marks on the floor and headphone trackers in the room corners

In addition, a hearing training was developed to familiarize the participants with the system and focus their senses on hearing. Initially, the six portraits of the inventors were covered by pictures of cats. The participants positioned themselves in the marked area in the middle of the room and were instructed that they would hear each of the cats meow and that they should point to the cat from which they perceived the meowing. If the participants identified all six cats correctly, they passed the hearing training. If the participants did not pass the hearing training the first time, the hearing training was repeated a maximum of two additional times. If the participants did not pass the hearing test after the third trial, they were excluded from data analysis.

Measures

Prior knowledge about inventors and their inventions

For each inventor and for each invention, prior knowledge was assessed in paper–pencil form by a six-point Likert scale ranging from 1 (no prior knowledge at all) to six (very high prior knowledge). Separate mean scores for inventors and inventions were calculated, each ranging from 1 to 6. Participants with values greater than 3 on either of these two scores were excluded from the analyses.

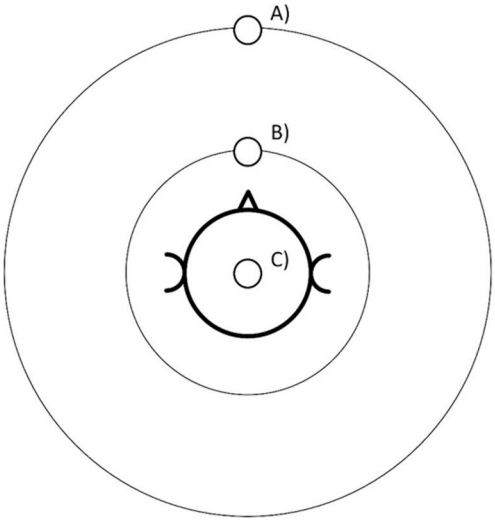
Audio quality

To examine the audio quality in both conditions, a paper–pencil questionnaire was generated. The first part of the questionnaire measured the participants' overall impression of the listening experience (overall impression), how well they were able to attribute a direction to the listening experience (localizability), their perception of how well the audio explanations and the corresponding portraits formed one unit (fusion), and their perception of how well the reverberation of the audio explanations fitted their expectations of the exhibition room (reverberation). Answers had to be given on Bodden-Jekosch scales (Köster et al., 2015) ranging from − 3 to + 3: − 3 (extremely bad), − 2 (bad), − 1 (meagre), 0 (fair), + 1 (well), + 2 (excellent), and + 3 (ideal) in steps of .2 points. These items were analyzed separately. The second part of the questionnaire examined sound externalization with a scale based on the work by Plenge (1972) described above and the binary ratings of externalization named by Best et al. (2020). Here, on a graphic depicting a head, a circle within the head, a close circle around the head, and a wide circle around the head, the participants had to mark one of the three circles to describe at which positions they perceived the audio explanations to have come from (see Fig. 4). Answers were coded with zero ("The listening experiences were predominantly inside my head."), one point ("The listening experiences were predominantly external but very close to my head.") or two points ("The listening experiences were predominantly outside my head in space."). This makes it possible to map the binary character of localization "in the head" (internalization) and "outside the head" (externalization) on a scale. At the same time, however, it also considers the fact that auditory events can also be perceived "close to the head", which, depending on the individual, can be rated as either inside or outside the head on a binary scale. These perceptual events are mapped with the category "close to the head" in the scale used. The last part of the questionnaire assessed possible artefacts of the audio presentation by single items (number of audio skips, number of jumps in the auditory event direction, number of jumps in distance perception, number of other possible artefacts that the participants were asked to write down). Each of these items had to be answered on a six-point Likert scale ranging from 1 to 6: 1 (none), 2 (very few), 3 (few), 4 (some), 5 (many), and 6 (very many) and was analyzed separately.

Judgement of learning

Exploratorily, judgement of learning was measured in paper–pencil form by one item which asked: "Please indicate on a scale from 0 to 100 (percent) how confident you

Please indicate whether you perceived the listening experiences more in your head or outside your head. In doing so, please note the information below on the individual evaluation options A, B and C and mark one of them with a cross.



A) Outer circle: "The auditory experiences were predominantly outside my head in space."

B) Middle circle: "The auditory experiences were predominantly external but very close to my head."

C) Center circle: "The listening experiences were predominantly inside my head."

Fig. 4 Item measuring externalization translated into English

are about correctly answering the questions in the following knowledge test about the exhibition content". Answers were given on a 10-point Likert scale ranging from 10% (very unconfident) to 20%, 30%, 40%, 50%, 60%, 70%, 80%, 90%, and 100% (very confident).

Memory of audio explanation details about inventors and their inventions

In research about cognitive processing and learning based on the CTML (Mayer & Fiorella, 2021), CATLM (Moreno & Mayer, 2007), and ICALM (Plass & Kaplan, 2016), memory of learning content is usually measured either by retention or recognition tests, whereby retention can be seen as the higher order cognitive process compared to recognition. This should also be appropriate for measuring learning based on our MAVCP. We therefore measured memory of audio explanation details about inventors and their inventions by a retention test in paper–pencil form with open questions – four questions about each of the six inventors and four questions about each auf the six inventions. Participants received one point if they recalled a detail correctly. Ratings were done by two independent raters. Interrater reliability (Krippendorff's alpha) was high with $\alpha = .96$ for items on inventors and $\alpha = .95$ for items on inventions. In cases in which these raters did not agree, a third rater made the decision. Separate proportion correct scores were calculated for recall of details about the inventors and for recall of details about the inventions.

Text-picture integration

Since learning based on the CTML (Mayer & Fiorella, 2021), CATLM (Moreno & Mayer, 2007), ICALM (Plass & Kaplan, 2016), and our MAVCP includes the integration of auditive or written texts and pictures as an important aspect of learning, a text-picture integration test in paper pencil form was used to assess how well participants were able to connect the details in the audio explanations with the corresponding portraits. Participants were shown portrait details depicting the face of each inventor together with four different statements about his invention (one correct and three incorrect) and were asked to mark the one correct statement. They received one point for each correctly answered item. A proportion-correct score was calculated across the six items.

Interest in further information about the inventors and their inventions

Exploratorily, interest in further information about the inventors and their inventions was measured in paper–pencil form with three questions about each inventor and three questions about each invention. Items had to be answered on a six-point Likert scale ranging from 1 (does not apply at all) to 6 (fully applies). Separate mean scores were calculated for interest in further information about the inventors and for interest in further information about the inventions across all six portraits.

Procedure

The participants took part in the experiment in individual sessions. They were welcomed, signed a consent form, and then filled out the questionnaire on demographics (age, gender, field of study), exclusion criteria (native speaker level, vision and hearing ability), and prior knowledge. Then they entered the exhibition room with the pictures of the cats covering the portraits of the inventors and their inventions. They were positioned in the marked area near the experimenter desk and put on the headphones. After checking the audio in this area, they were instructed to position themselves in the marked area in the middle of the room to do the hearing training. After completing the hearing training, the experimenter took off the pictures of the cats, revealing the portraits of the inventors and their inventions. The participants were then instructed to navigate through the exhibition in the order marked on the floor and to listen fully to the automatically played audio explanations in front of the respective portraits. They were also instructed to move freely within the marked areas while listening, that turning their head could help with listening, and that after the exhibition visit, they will be asked questions about it. The audio explanations were played either in spatial or normal audio mode, depending on the condition the participants were in. After the exhibition visit, the participants left the exhibition room and were led into an adjacent room to answer the questionnaire about the audio quality. Next, they worked on an intermediate task for 8 min—a memory puzzle in which word-picture pairs had to be found—to override information in working memory. They then answered the judgement of learning item, the memory test about audio explanation details on inventors and their inventions, the text-picture integration test, and the questionnaire about their interest in further information about the inventors and their inventions. Finally, the

participants were thanked and paid 12,50 euros for their participation. Each study session lasted approximately 75 min.

Results

For all dependent variables that were not normally distributed within the single conditions, non-parametric Mann–Whitney *U*-tests instead of independent *t*-Tests were calculated. With regard to the dependent variables related to the hypotheses (memory of audio explanation details about inventors, memory of audio explanation details about inventions, text-picture integration), one-tailed tests were calculated. Regarding the exploratory variables (interest, audio quality, judgement of learning), two-tailed tests were calculated.

Audio quality

Regarding overall impression, localizability, fusion, and reverberation, no significant differences were found between the two conditions. Mean values, standard deviations, and test statistics for the four items are presented in Table 1.

The item on externalization showed significant differences between the conditions, $\chi^2(2) = 11.34$, $p = .003$, $\phi = .49$. With the spatial audio settings, four participants answered that they perceived the audio explanations external in the room, 20 participants external near the head, and one participant internal in the head. With the normal audio settings, seven participants answered that they perceived the audio explanations external in the room, eight participants external near the head, and eight participants internal in the head.

For the number of different possible artefacts (audio skips, jumps in the auditory event direction, jumps in distance perception), no significant differences were found. Mean values, standard deviations, and test statistics for the three items are presented in Table 2.

Judgement of learning

There was no significant difference in judgement of learning between spatial audio settings ($M = 48.00$, $SD = 13.84$) and normal audio settings ($M = 43.48$, $SD = 16.95$), $U = 241.50$, $p = .334$, $r = .14$.

Table 1 Mean values, standard deviations, and test statistics with regard to four dimensions of audio quality in the conditions with spatial and normal audio settings

	Spatial audio		Normal audio		*t*(*df*)	*U*	*p*	Cohen's *d*	*r*
	M	*SD*	*M*	*SD*					
Overall impression	1.92	.65	1.61	.64		211.50	.107		.23
Localizability	.86	1.32	1.26	1.11	1.12(46)		.268	.32	
Fusion	1.22	.91	1.23	.90		283.00	.924		.01
Reverberation	1.26	.96	1.21	.85	.21(46)		.834	.06	

Table 2 Mean values, standard deviations, and test statistics with regard to the number of possible artefacts in the conditions with spatial and normal audio settings

	Spatial audio		Normal audio		*U*	*p*	*r*
	M	*SD*	*M*	*SD*			
Audio skips	1.24	.44	1.35	.49	318.50	.417	.12
Jumps in the auditory event direction	1.56	.82	1.17	.49	218.50	.060	.27
Jumps in distance perception	1.28	.54	1.39	.66	308.50	.577	.08

Regarding the item about the number of other possible artefacts, from the $n=48$ participants, answers on the scale were given by $n=25$ (52.08%) participants and $n=10$ (20.83%) participants specified these artefacts in written form, for example, crackling, popping, and noise. Since the scale values refer to different artefacts, no inferential statistics were calculated for this item

Memory of audio explanation details about inventors and their inventions

There was a significant difference in memory of audio explanation details between the two conditions but only for the audio explanation details about the inventions, $U=194.00$, $p=.027$, $r=.28$, and not for the audio explanation details about the inventors, $t(46)=.69$, $p=.247$, $d=1.99$. While details about the inventions were recalled significantly better with spatial audio settings ($M=.46$, $SD=.22$) than with normal audio settings ($M=.34$, $SD=.20$), recall of details about the inventors was similar with spatial audio settings ($M=.45$, $SD=.12$) and with normal audio settings ($M=.43$, $SD=.14$).

Text-picture integration

Regarding text-picture integration, the results showed no significant differences between spatial audio settings ($M=.74$, $SD=.22$) and normal audio settings ($M=.63$, $SD=.31$), $U=237.50$, $p=.145$, $r=.15$.

Interest in further information about the inventors and their inventions

No significant differences between the conditions regarding interest in further information about the inventors, $t(46)=-.50$, $p=.623$, $d=.14$, and their inventions, $t(46)=.20$, $p=.844$, $d=.06$, were found. The participants with spatial audio settings were similarly interested in further information about the inventors ($M=2.56$, $SD=1.00$) and the inventions ($M=3.00$, $SD=1.19$) as those participants with normal audio settings (inventors: $M=2.72$, $SD=1.24$; inventions: $M=2.89$, $SD=1.17$).

Discussion

By comparing the use of a spatial versus a normal audio guide in an experimental exhibition, Experiment 1 examined the effects of audio-visuo-spatial contiguity on learning. The results showed that memory of audio explanation details about the inventions was better with spatial than with normal audio (a medium instead of the expected large effect and only statistically

significant in one-sided testing, indicating that more participants would have been needed to power this effect), whereas memory of audio explanation details about the inventors did not differ, partially confirming H1. In contrast, the results showed no significant differences between spatial and normal audio guides regarding text-picture-integration, not confirming H2. The results further showed that the impression of externalization differed significantly between both conditions, indicating that the experimental manipulation was reflected by the participants' auditory perception. More specifically, in line with our expectations, audio explanations in the spatial audio condition were rated to be located mainly external near the head. However, in the condition with normal audio, the impression of externalization was distributed across all possible answers instead of being perceived mainly inside the head as expected, even though all participants had successfully passed the spatial hearing training. A possible explanation that a substantial proportion of the participants in the normal audio condition did locate the audio explanation outside their head is given by Chion (1994, cited in Terrenghi & Zimmermann, 2004). According to him „people possess a natural urge to merge sounds and images into a cohesive whole as a strategy for making sense of the world" (Terrenghi & Zimmermann, 2004). In the present experiment, this tendency was fostered by the fact that the inventors in the portraits looked directly at the participants and that the audio explanations were formulated in first- and second-person perspective directly addressing the participants. Taken the results on memory together with the results on externalization, not only audio-visuo-spatial contiguity, but also externalization may have played a role for memorizing the learning content presented. From an evolutionary-biological perspective, external compared to internal perceptions of audio information are the more natural ones that may have led to faster orientation and deeper processing and therefore better memory.

Audio-spatial cueing may also provide an explanation for the present findings. For example, Baldis (2001) found that contributions in a desktop conference were better matched with conference members if they were presented via different loudspeakers, each assigned to one of the members, than by a single loudspeaker. This effect of audio-spatial cues is in line with the context maintenance and retrieval model of memory search (CMR), which postulates that learning content is encoded together with stimulus-related and source-related context features (Polyn et al., 2009). During memory retrieval of particular information, the context in which it was learned is reactivated, making other content sharing the same context more accessible. This is corroborated by studies of episodic memory (e.g., Clark & Bruno, 2021), priming (e.g., Clayton & Habibi, 1991), and memory of audio-visual information presentations (e.g., Miller et al., 2013). Accordingly, spatial locations of learning content constitute one important source-related context feature. In Experiment 1, in the spatial audio condition, audio explanations provided such spatial cues because each explanation was located within a different portrait. In the normal audio condition, audio explanations were all located within the head of the person, thus providing no different spatial cues. However, since the person moved from portrait to portrait through the exhibition room, the different audio explanations were related to different standpoints and therefore provide different spatial cues for the different audio explanations in the normal audio condition.

Experiment 2

The aim of Experiment 2 was to disentangle audio-visual-spatial contiguity, externalization, and audio-spatial cueing as possible explanations of the results in Experiment 1. We realized three conditions by keeping sound externalization constant and

manipulating audio-visuo-spatial contiguity and audio-spatial cueing as follows: presenting the auditory information either at the position of the corresponding exhibition object (condition close, with audio-spatial cues), spatially separated and lying on the opposite wall of the corresponding exhibition object (condition distant I, with audio-spatial cues), or for all exhibition objects at the same position in space (condition distant II, without audio-spatial cues). The following preregistered hypotheses were postulated:

H1 In the case of an audio-visuo-spatial contiguity effect, the audio explanation content should be remembered better when the audio explanations are perceptually located within the portraits (close), but not spatially separated from them (distant I, distant II).

H2 In the case of an audio-spatial cueing effect, the audio explanation content should be remembered better when each audio explanation is presented at a different position in space and thus provides its own spatial cues (close, distant I), but not when all audio explanations are presented at the same position in space and thus provide the same spatial cue (distant II).

Additionally, externalization and perceived direction of the sound source of all audio explanations as well as perceived plausibility of the audio-room congruency were measured as control variables.

Method

Participants

Data were collected from a total of 171 participants. According to previously defined exclusion criteria (also used in Experiment 1), 25 participants were excluded from the analyses. In addition, 23 participants who failed to pass a manipulation check (description see below) were also excluded from the analyses. Hence, a total sample of $n = 123$ participants were analyzed. This is exactly the number of participants needed according to an a priori power analysis with G*Power calculated to find a medium effect of $d = .57$ (effect size of Experiment 1) with a power of .80 and an alpha error probability of .05. The participants were aged between 18 and 48 years ($M = 24.67$, $SD = 4.66$). They were predominantly students from different fields of study (e.g., law, geography, cognitive science, psychology, English) and consisted of 97 females, 24 males and two diverse participants. A positive ethics vote of the local ethics committee was obtained for the study.

Design

We used a one-factorial design with the audio location condition (audio explanations located within the corresponding portraits (close), behind the participants at the opposite wall (distant I), or from one and the same position in the room, which was on the wall opposite to the first portrait (distant II)) as the between-subjects factor. Figure 5 illustrates the perceived spatial positions of the sound sources for each of the three conditions.

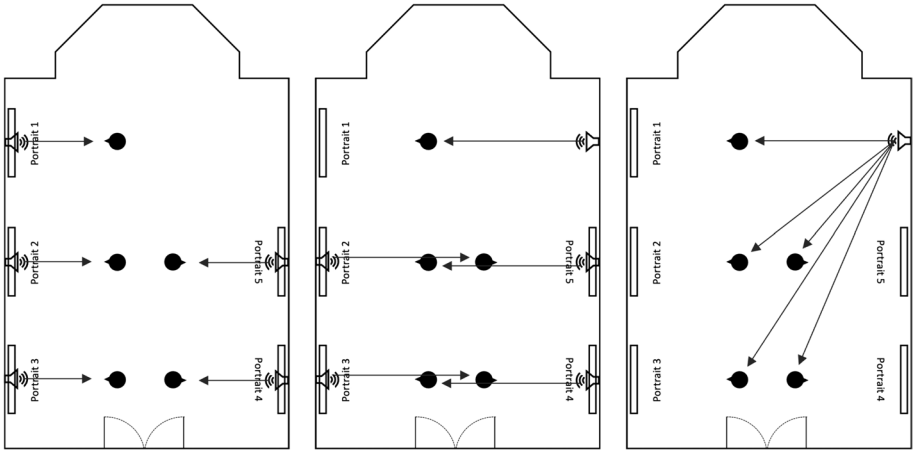


Fig. 5 Experimental conditions varying in the perceived location of audio explanations. A distinction is made between audio explanations always perceived from the picture (left), always perceived from the back (center), and always perceived from the same spatial position (right). The black dots represent the respective position and viewing direction of the participant in front of each portrait

Material

The same portraits and audio explanations as in Experiment 1 were used except the ones about Douglas Engelbart (computer mouse) because, due to technical reasons, one of the audio positions of Experiment 1 had to be used as the portrait-independent sound source in the condition with audio explanations perceived from a single place in the room (distant II). The audio explanations were presented with the same audio system as in Experiment 1. The spatial setup of the marked areas was the same as in Experiment 1, with one exception: Compared to Experiment 1, the distance between the portraits and the areas in which the participants should stand in front of the portraits to listen to the respective audio explanations was doubled to a distance of 190 cm, leading to a reduction in the size of the area which was now a 190 × 50 cm wide rectangular.

Measures

Prior knowledge

As an exclusion criterium, prior knowledge was measured by the same instrument as in Experiment 1, except that the items about Douglas Engelbart and the computer mouse were omitted. Again, separate mean scores for inventors and inventions were calculated, each ranging from 1 to 6. Participants with values greater than 3 on either of these two scores were excluded from the analyses.

Manipulation check

As an exclusion criterium, we checked whether the participants actually perceived the audio explanations to be coming from the intended positions according to the respective conditions by the use of a single-choice question in paper–pencil format. The participants were asked to answer the question "Where did the sound regarding the five pictures you just looked at come from?" from the three answer options "always from the picture in front of me", "always from the picture behind me" and "always from the same place in the room". The answer options corresponded the audio settings in the three conditions. Participants who were not able to answer this question according to their experimental condition were excluded from the analyses.

Externalization

Perceived externalization was measured with a paper–pencil questionnaire. First, a sketch of the previously visited exhibition space was presented to remind the participants of the spatial layout of the previously visited exhibition. Then they were instructed to fill out the externalization item used in Experiment 1, separately for each audio explanation. Contrary to Experiment 1, participants were not only instructed to mark their perceived externalization on circles at different distances around a simplified representations of the participant's head but also to place their cross at the respective direction from which they perceived the audio explanations. Externalization was coded as in Experiment 1, and for each participant, a mean value of externalization was calculated across the five audio explanations. In addition, a score for the direction from which the audio explanations were perceived to be emanating was calculated. In the case of a sound source lying in the portrait being viewed, the mark was to be placed in front of the head, and in the case of a sound source lying on the rear wall, the mark was to be placed accordingly on the rear side of the head. In the condition with all audio explanations coming from the same position in the room, the mark was to be placed at different degrees on the depicted circles depending on the portrait in question. The markings were evaluated by use of stencils. Markings that were within the tolerance range of the stencils were coded with one point and markings outside the tolerance range of the stencil were coded with zero points. For each participant, a proportion correct score was calculated across the five different audio explanations.

Plausibility of audio-room congruency

The perceived fit between the heard acoustic properties of the audio explanations and the acoustic properties of the exhibition space was measured with a single paper pencil item ("How much did it sound to you like the inventors were actually standing in the room with you, or how well did the acoustical properties of the audio information that you heard match the acoustical properties of the exhibit space?") including a Bodden and Jekosch scale (Köster et al., 2015)—a .2-steps rating scale ranging from − 3 to + 3: − 3 (extremely bad), − 2 (bad), − 1 (meagre), 0 (fair), + 1 (well), and + 2 (excellent) to + 3 (ideal).

Memory of audio explanation details about inventors and their inventions

Memory of audio explanation details about inventors and their inventions was measured as in Experiment 1, except that the items about Douglas Engelbart and the computer mouse were omitted. The responses of the participants were again rated by two independent raters. In the case of discrepant ratings, a third rater made the final decision as to whether an item was answered correctly or not. Interrater reliability coefficients (Krippendorff's alpha) with $\alpha = .87$ for items about inventors and $\alpha = .89$ for items about inventions indicated high agreement between the two raters. Again, for both the items about inventors and the items about inventions separate proportion-correct scores were calculated.

Text-picture integration

Text-picture integration was measured as in Experiment 1, except that the item about Douglas Engelbart (computer mouse) was omitted. Again, a proportion-correct score was calculated.

Procedure

The procedure of the experiment was the same as in Experiment 1 with the following exceptions: To further encourage the participant's head movements to better perceive the differences in the audio presentations between the conditions, the participants were asked to turn toward the experimenter at the beginning of each audio explanation to signal whether the audio explanation could be heard well. Further, after the exhibition visit, the participants first completed the manipulation check and then answered the externalization questionnaire and the item about the perceived plausibility of the audio-room congruency instead of a more general audio quality questionnaire. After the intermediate task, the memory and text-picture integration tests were administered. Finally, the participants were asked whether they had already known the exhibition content.

Results

According to Shapiro–Wilk tests, externalization, perceived direction, plausibility of audio-room congruency, memory of audio explanation details about inventors, and text-picture integration were not normally distributed within the single conditions. Therefore, the differences between the conditions regarding these variables were calculated by Kruskal–Wallis tests. The differences between conditions in the normally distributed variable memory of audio explanation details about inventions were calculated by a one-factorial analysis of variance (ANOVA). One-tailed analyses were applied to the hypotheses-related dependent variables memory of audio explanation details about inventors, memory of audio explanation details about inventions, and text-picture integration. Two-tailed analyses were applied to the control variables externalization, perceived direction, and plausibility of audio-room congruency.

Externalization

A Kruskal–Wallis test showed no significant differences of externalization between the conditions, $H(2) = 4.05$, $p = .132$, $r = .37$. As expected, externalization was perceived to be similar in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = 1.11$, $SD = .46$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I; $M = 1.28$, SD = ..39), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = 1.31$, $SD = .37$).

Also, no significant differences regarding the perceived direction of the audio explanations were found between the conditions, $H(2) = 1.15$, $p = .563$, $r = .10$. The directions of the audio explanations were identified similarly well in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = .48$, $SD = .34$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I; $M = .46$, $SD = .34$), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = .40$, $SD = .31$). Note that in this analysis, one of the participants in the close condition could not be included due to missing values.

Plausibility of audio-room congruency

A Kruskal–Wallis test showed no significant differences between the conditions, $H(2) = 2.97$, $p = .227$, $r = .27$. Audio-room congruency was perceived to be similarly plausible by the participants in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = 1.19$, $SD = .76$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I; $M = .75$, $SD = 1.25$), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = .89$, $SD = 1.25$).

Memory for audio explanation details about inventors and their inventions

A Kruskal–Wallis test regarding the memory of audio explanation details about inventors showed no significant differences between the conditions, $H(2) = .51$, $p = .389$, $r = .05$. Memory of audio explanation details about the inventors was similar in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = .45$, $SD = .13$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I; $M = .46$, $SD = .15$), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = .44$, $SD = .13$).

A one-factorial analysis of variance (ANOVA) regarding the memory of the audio explanation details about inventions, too, showed no significant differences between the conditions, $F(2, 120) = .34$, $p = .358$, $\eta_p^2 = .01$. Memory of audio explanation details about the inventions was similar in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = .41$, $SD = .20$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I;

$M = .44$, $SD = .19$), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = .44$, $SD = .20$).

Text-picture integration

A Kruskal–Wallis test showed no significant difference between the conditions, $H(2) = .97$, $p = .308$, $r = .09$. Text-picture integration was similar in the condition with audio explanations perceived to be located within the corresponding portraits (close; $M = .74$, $SD = .23$), in the condition with audio explanations perceived behind the participants at the opposite wall (distant I; $M = .77$, $SD = .20$), and in the condition with audio explanations perceived from one and the same position in the room (distant II; $M = .72$, $SD = .24$).

Discussion

Experiment 2 aimed to disentangle three possible mechanisms responsible for the positive memory effect of perceptually locating corresponding audio explanations within the portraits compared to within the headphones that was shown in Experiment 1, namely audio-visual-spatial contiguity, externalization, and audio-spatial cueing. For this purpose, visitors' memory of audio explanations and text-picture integration were compared when audio explanations were perceived to be located within the corresponding portraits (close), behind the participants at the opposite wall (distant I), or from one and the same position in the room which was on the wall opposite to the first portrait (distant II).

In line with Experiment 1, no significant effects between the conditions were found for both memory of audio explanation details about the inventors and for text-picture integration. Contrary to Experiment 1, also no significant differences between conditions were found for memory of audio explanation details about the inventions. Thus, neither of the two hypotheses could be confirmed regarding the memory tests. However, in all three

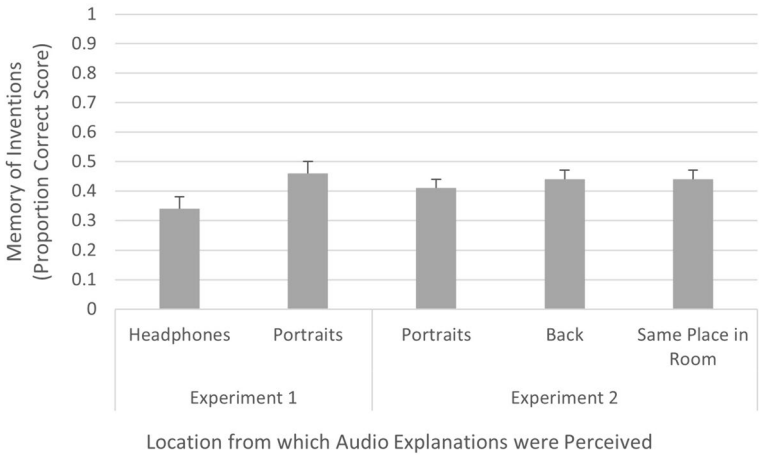


Fig. 6 Mean values and standard errors of the memory of audio explanation details about inventions for the experimental conditions in Experiment 1 and Experiment 2

conditions, mean values for memory of audio explanation details about the inventions were descriptively similar to the mean values of Experiment 1's condition with audio explanations located within the corresponding portraits (close), but not to the condition with audio explanations located within the headphones (see Fig. 6). Hence the pattern of results tentatively suggests that neither spatial contiguity nor spatial cueing, but instead, externalization may be an explanation for the positive memory effects found in Experiment 1. However, it is always problematic to infer the absence of an effect from non-significant results. Hence, our current results certainly cannot make any final statements about the underlying mechanism of a possible beneficial memory effect of spatial audio guides but only give a first indication that it might be an externalization effect. This has to be confirmed in further studies.

General discussion

The present work examined effects of spatial audio guides on learning in informal spatial multimedia learning settings and the possible mechanisms behind such effects. In a first experiment, we referred to the MAVCP and the spatial contiguity principle (Fiorella & Mayer, 2021) deduced from CTML (Mayer & Fiorella, 2021) to hypothesize an advantage of spatial audio guides over normal audio guides on memory performance, thereby generalizing from previous two-dimensional visual learning material to learning content arranged in the three-dimensional space of an exhibition—in our case, an exhibition about portraits of inventors and their inventions. The results showed a beneficial effect of spatial audio guides on memory for information about the presented inventions, whereas no benefit was found for information about the inventors and for text-picture-integration, partly confirming the assumption of a learning advantage of spatial audio over normal audio presentations.

However, differences in perceived externalization between the two conditions in Experiment 1 indicated that this effect could not only be caused by spatial contiguity but also by sound externalization. Perceiving the audio explanations to be coming from somewhere external and not internal in the headphones or from one's own head may be—from an evolutionary biology perspective—a more natural condition that affords less cognitive load and facilitates learning. This is in line with the concept of biological primary knowledge, that is, knowledge, information, and skills that have implicitly evolved over many generations, that impose low load on working memory, and therefore have a positive influence on learning biological secondary knowledge—cultural knowledge that has to be taught (Kirschner et al., 2018; Paas & Sweller, 2012). Processing information from external sound sources may be regarded as biological primary knowledge, while processing information presented via headphones may not. This, too, may explain why learning with spatial compared to normal audio guides was better.

Another explanation for the beneficial effects of spatial audio guides on the memory performance of the visitors may be that by using spatial audio guides, different location information is linked to the different audio explanations and that these audio-spatial cues fostered retrieval of the learning content (cf., Baldis, 2001).

To distinguish between spatial contiguity, externalization, and audio-spatial cueing as possible mechanisms explaining the beneficial effects of spatial audio guides on memory, a second experiment was conducted. It showed no differences in memory performance between audio explanations perceptually located within the portraits and audio

explanations perceptually located separated from the corresponding portraits, thus providing no evidence for the audio-visuo-spatial contiguity effect. Unlike several studies with two-dimensional visual learning material (cf. Ginns, 2006; Schroeder & Cenkci, 2018), the spatially close presentation of audio-visual learning material in the exhibition setting used here appeared not to save processing costs and therefore enhance learning. This is in line with Krüger and Bodemer (2022) and Liu et al. (2012), who also did not find positive effects on learning performance when learning content was presented spatially integrated compared to separated. Contrary to these studies, we used audio texts instead of written texts. The absence of a spatial contiguity effect with audio-visual learning content may indicate that spatial contiguity effects are based on cognitive costs due to visual search processes and not to spatial contiguity per se. Nevertheless, further studies on (audio-visuo-) spatial contiguity in spatial learning scenarios are needed to come to robust conclusions in this regard. At the same time, cognitive load as an indicator of cognitive processes during the learning experience should also be assessed.

Experiment 2 also showed no differences in memory performance between audio explanations perceptually located at different positions in space and at the same position in space, providing no evidence for an audio-spatial cueing effect. It seems that the audio-spatial cues associated with the different perceived locations of the audio sources were not effective enough to foster the recall of the learning content as was assumed based on theories in the field of episodic memory research (Polyn et al., 2009) and empirical evidence that spatial cues play an important role in memory retrieval in spatial learning settings (Clark & Bruno, 2021; Clayton & Habibi, 1991; Miller et al., 2013). This may be because the retrieval of information is not only organized by audio-spatial but also by visuo-spatial, temporal, or semantic cues and because spatially structured retrieval depends on the type of learning task, the type of retrieval task, and which of these cues are used (e.g., Clark & Bruno, 2021). In the present learning setting, participants were required to link successively viewed portraits of inventors and their inventions to corresponding audio explanations. Therefore, visuo-spatial, temporal, and semantic cues may have possibly outweighed the effects of audio-spatial cues.

Finally, since externalization was kept constant across all conditions of Experiment 2 and because memory performance in all conditions was descriptively as high as in the condition with audio explanations perceptually located as external within the portraits in Experiment 1, the results suggest that externalization is the most probable mechanism to explain the beneficial memory effects of spatial audio guides found in Experiment 1. However, since this conclusion is based on null-effects, further studies are needed to support it, for example, by realizing all conditions in one study, together with a plausibility measure to check whether externalized perception of sound sources is perceived as more natural compared to the perception of sound sources in the head(phones).

Limitations

Why this effect of spatial audio guide on memory occurred only for the information about the inventions and not the inventors is unclear. It cannot be due to participants assuming that learning content in the exhibition was mainly about the inventions and not so much about the inventors and that they therefore generally ignored information about the inventors because mean values of memory of audio explanation details about inventors in both conditions were clearly above floor level. Future studies may therefore systematically examine this content-dependency.

One reason why there was no significant difference between spatial and normal audio guides regarding text-picture-integration may be due to the learning material. While spatial contiguity effects occur especially when the related textual and visual information can only be understood in mutual dependency (Ayres & Sweller, 2021; Mayer & Fiorella, 2014), the audio explanations in our studies may have been comprehensible without viewing the portraits. Future studies should therefore examine the present effect with learning content that is explicitly designed to require text-picture integration. Such learning content, for example, could be historical paintings. For these, audio explanations could provide a description and interpretation of the artwork and the events depicted, which may encourage the visitors to recognize and follow the auditorily presented information in the artwork itself (Glaser et al., 2023).

Another factor that may have influenced the findings is the small size of the room in which the experimental study was set up. A larger room could be used in which clearer differences in distances between visitors and perceived audio sources and therefore different degrees of externalization could be realized to examine possible externalization dependent memory effects. In addition, using a more sensitive externalization measure to measure differences in externalization caused by different distances between the visitors and the perceived audio sources in the different conditions might be advisable. One such measure could be a continuous scale in percentages (cf. Leclère et al., 2019) with predefined directions.

Practical implications

Besides contributing to a better understanding of the above-described theories about the cognitive processing of multimedia learning material, (audio-visuo-) spatial contiguity, and (audio-)spatial cueing as well as shedding more light onto the phenomenon and consequences of sound externalization, there are also some practical implications that can be drawn from the present experiments. First, in view of the results presented here, the externalized perception of audio information by spatial audio guides might generally have a more positive effect on learning performance than normal audio guides lacking such externalization. However, many participants were unable to correctly assign the positions of the presented virtual sound sources in the auditory training and the manipulation check. This indicates that the use of spatial audio guides can cause difficulties in localizing the sound source, which is in line with the fact that spatial localization of auditory stimuli is less dominant in humans than spatial localization of visual stimuli (Alais & Burr, 2004). In addition, a spatial audio guide in its technical implementation is not specifically adapted to the visitors' individual hearing characteristics but to the measurements on an artificial head. For this reason, the auditory impression obtained by a spatial audio guide with headphones does not necessarily correspond to that which a visitor would have in the presence of real instead of virtual sound sources (Stitt et al., 2019). However, in real museums and exhibitions, visitors may benefit from moving more freely within a large space to experience the auditory impression in different orientations and stance positions. Under such conditions, a more active engagement with the spatial audio guide can be encouraged and visitors may fully exploit the possibilities that such a system offers in terms of its acoustic properties (cf. Zimmermann & Lorenz, 2008). This could even be enhanced by designing the spatial audio guide in such a way that visitors are "called" by exhibition objects throughout the space, drawing them in to engage with them in greater depth. To make a sound recommendation for the design of spatial audio guides in museums or exhibitions,

empirical studies of the effects of different designs of spatial audio guides on learning are needed—not only in the laboratory but also in the field.

Outlook

As mentioned above, the next research steps should be to examine (audio-visuo-) spatial contiguity, externalizations effects, and their content dependency in spatial learning scenarios more closely, for example, by combining the conditions of our two experiments in a single experiment. Such future studies should also include considering cognitive load, perceived plausibility of the externalized perception of sound sources, and room size as influencing factors. This would result in the clarification of the underlying psychological mechanism of the possible learning benefits of spatial compared to normal audio guides and contribute to the empirical validation of the MAVCP.

Further, to determine boundary conditions of spatial audio guides on learning, the effects should be examined with different types of exhibits such as artistic paintings of various styles, historical paintings, sculptures, historical remains, or devices. Going beyond exhibition settings, different types of learning scenarios—informal ones (e.g., cultural heritage sites, zoos, or aquariums) and formal ones (e.g., online group discussions and teamwork, virtual reality trainings)—should be examined.

To enhance the applicability of spatial audio guides for learning purposes, further research should develop optimized spatial audio guides and overcome the current disadvantages such as the above-mentioned difficulties of learners to localize externalized sound sources and the suboptimal adaption of the technical implementation of current spatial audio guides to individual hearing characteristics. Finally, field studies should examine the usability and learning with spatial audio guides in order to detect practical challenges, develop optimized systems, and empirically confirm the importance of (audio-visuo-) spatial learning in different complex, spatial multimedia learning scenarios.

Future exhibitions, for example, may then focus less on the visual sense, but more on including types of object presentations that address multiple senses and especially the auditory sense. Exhibition narratives, for example, could be included in the exhibitions more naturally by letting the objects themselves tell their story by speaking directly to the visitors from their individual positions to the respective positions of the visitors instead of telling the exhibition narrative via normal audio guides or text panels. In this way, the museum experience may become a more natural, entertaining, and engaging learning experience.

Conclusions

The presented experiments provide evidence for a beneficial effect of spatial audio guides on learning in spatial multimedia learning settings, such as museums and exhibitions. Although there was no clear evidence for a particular mechanism explaining this effect, the results indicate that externalization rather than audio-visuo-spatial congruency and audio-spatial cueing may be responsible. The experiments therefore provide a starting point for further research examining learning with spatial audio guides. Furthermore, the experiments provide valuable insights into practical aspects that can be used for further research and the use and design of spatial audio guides in museums and exhibitions. There may be a great potential of spatial audio guides for learning, based on more research to clarify the boundary conditions and cognitive processes that enable fully exploiting these potentials.

Appendix

Transcript of the audio text on Eduard Justi and his continuous current electromagnet

Allow me to introduce myself. My name is Eduard Wilhelm Leonhard Justi. I was born in Hong Kong in 1904 to German parents and became known as a physicist and university teacher. I studied physics, chemistry, mathematics, and geology at the universities of Marburg, Kiel, and Berlin. There I learned from the most famous physicists of our time: Albert Einstein, Max Planck, and Erwin Schrödinger. From 1929 to 1944, I worked at the Physikalisch-Technische Reichsanstalt of Berlin. There I was interested, among other things, in low-temperature physics, which deals with physical processes in very cold matter.

In my hand, I hold parts of my continuous current electromagnet. First, let me explain what an electromagnet is all about: An electromagnet usually consists of one or more coils, which are wound metal wires through which an electric current flows. Since each winding of the coil acts like a circular conductor, a magnetic field is created for each winding. These magnetic fields then overlay each other to create a large overall magnetic field. We can often find an iron core in the center of a coil, which further enhances the magnetic field. I have combined the principle of the electromagnet with the phenomenon of the so-called "superconductivity". A superconductor is a substance or metal that no longer has any electrical resistance when the temperature falls below a certain level. The temperature to be fallen below is also called "transition temperature". In my experiment, I used a winding made of the superconductor niobium nitride and cooled it down using liquid hydrogen. Even before the niobium nitride ring reaches its transition temperature and thus no longer has any resistance, I switch on a coil underneath it. This creates a magnetic field between the coil and the niobium nitride ring, which causes current to flow in the niobium nitride ring. If I now cool the ring further down to its transition temperature and switch the coil off again, a continuous current flows in my niobium nitride ring. A continuous current is thus an electric current that can flow through a superconductor on a self-contained path without loss and at a constant rate.

Thus, I discovered that the current continues to flow unimpeded for several hours at a constant low temperature of hydrogen due to the superconducting property of the niobium nitride ring. In 1942, I was able to demonstrate to the press the continuous current electromagnet operated in this way, which in its present form is used, for example, in electric motors.

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Declarations

Conflict of interest The authors have no relevant financial or non-financial interests to disclose.

Ethical approval Approval was obtained from the ethics committee of the Leibniz-Institut für Wissensmedien, Tübingen, Germany. The procedures used in this study adhere to the tenets of the Declaration of Helsinki.

Informed consent Informed consent to participate in the studies was obtained from all individual participants included in the studies. Participants signed informed consent regarding publishing their data anonymously.

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References

Aitamurto, T., Boin, J. B., Chen, K., Cherif, A., & Shridhar, S. (2018). The impact of augmented reality on art engagement: liking, impression of learning, and distraction. In J. Y. C. Chen & G. Fragomeni (Eds.), *Proceedings of the 10th International Conference Virtual, Augmented and Mixed Reality: Applications in Health, Cultural Heritage, and Industry (VAMR) 2018, Las Vegas, NV, USA* (pp. 153–171). Springer. https://doi.org/10.1007/978-3-319-91584-5_13

Alais, D., & Burr, D. (2004). The ventriloquist effect results from near-optimal bimodal integration. *Current Biology, 14*(3), 257–262. https://doi.org/10.1016/j.cub.2004.01.029

Ayres, P., & Sweller, J. (2021). The split-attention principle in multimedia learning. In R. Mayer & L. Fiorella (Eds.), *The Cambridge handbook of multimedia learning* (pp. 199–211). Cambridge University Press. https://doi.org/10.1017/9781108894333.020

Baldis, J. J. (2001). Effects of spatial audio on memory, comprehension, and preference during desktop conferences. In *Proceedings of the SIGCHI conference on Human factors in computing systems* (pp. 166–173). https://doi.org/10.1145/365024.365092

Bauhoff, V., Huff, M., & Schwan, S. (2012). Distance matters: Spatial contiguity effects as trade-off between gaze switches and memory load. *Applied Cognitive Psychology, 26*, 863–871. https://doi.org/10.1002/acp.2887

Best, V., Baumgartner, R., Lavandier, M., Majdak, P., & Kopčo, N. (2020). Sound externalization: A review of recent research. *Trends in Hearing*. https://doi.org/10.1177/2331216520948390

Bregman, A. S. (1990). Auditory scene analysis. *The Perceptual Organization of Sound. Cambridge: MIT Press.* https://doi.org/10.7551/mitpress/1486.001.0001

Cammeraat, S., Rop, G., & de Koning, B. B. (2020). The influence of spatial distance and signaling on the split-attention effect. *Computers in Human Behavior, 105*, 106203. https://doi.org/10.1016/j.chb.2019.106203

Chion, M. (1994). *Audio-vision: Sound on screen*. Columbia University Press.

Clark, D. P., & Bruno, D. (2021). Time is of the essence: Exploring temporal and spatial organisation in episodic memory. *Quarterly Journal of Experimental Psychology, 74*(8), 1406–1417. https://doi.org/10.1177/1747021821993823

Clayton, K., & Habibi, A. (1991). Contribution of temporal contiguity to the spatial priming effect. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 17*(2), 263–271. https://doi.org/10.1037/0278-7393.17.2.263

D'Auria, D., Di Mauro, D., Calandra, D. M., & Cutugno, F. (2015). A 3D audio augmented reality system for a cultural heritage management and fruition. *Journal of Digital Information Management, 13*(4), 203–209.

Fiorella & Mayer (2021). Principles for Reducing Extraneous Processing in Multimedia Learning. Coherence, Signaling, Redundancy, Spatial Contiguity, and Temporal Contiguity Principles. In R. E. Mayer

& L. Forella (eds.), *The Cambridge Handbook of Multimedia Learning* (pp. 185-198). Cambridge University Press. https://doi.org/10.1017/9781108894333.019

Giariskanis, F., Kritikos, Y., Protopapadaki, E., Papanastasiou, A., Papadopoulou, E., & Mania, K. (2022). The augmented museum: a multimodal, game-based, augmented reality narrative for cultural heritage. In *ACM International Conference on Interactive Media Experiences* (pp. 281–286). https://doi.org/10.1145/3505284.3532967

Ginns, P. (2006). Integrating information: A meta-analysis of the spatial contiguity and temporal contiguity effects. *Learning and Instruction, 16*(6), 511–525. https://doi.org/10.1016/j.learninstruc.2006.10.001

Glaser, M., Knoos, M., & Schwan, S. (2023). How verbal cues help to see and understand art. *Psychology of Aesthetics, Creativity, and the Arts, 17*(3), 278–293. https://doi.org/10.1037/aca0000372

Grüninger, R., Specht, I., Lewalter, D., & Schnotz, W. (2014). Fragile knowledge and conflicting evidence: What effects do contiguity and personal characteristics of museum visitors have on their processing depth? *European Journal of Psychology of Education, 29*, 215–238. https://doi.org/10.1007/s10212-013-0195-0

Heller, F., & Borchers, J. (2015). Audioscope: smartphones as directional microphones in mobile audio augmented reality systems. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (pp. 949–952). https://doi.org/10.1145/2702123.2702159

Holsanova, J., Holmberg, N., & Holmqvist, K. (2009). Reading information graphics: The role of spatial contiguity and dual attentional guidance. *Applied Cognitive Psychology, 23*(9), 1215–1226. https://doi.org/10.1002/acp.1525

Jee, B. D., & Anggoro, F. K. (2021). Designing exhibits to support relational learning in a science museum. *Frontiers in Psychology, 12*, 636030. https://doi.org/10.3389/fpsyg.2021.636030

Johnson, C. I., & Mayer, R. E. (2012). An eye movement analysis of the spatial contiguity effect in multimedia learning. *Journal of Experimental Psychology: Applied, 18*(2), 178–191. https://doi.org/10.1037/a0026923

Kaghat, F. Z., Azough, A., Fakhour, M., & Meknassi, M. (2020). A new audio augmented reality interaction and adaptation model for museum visits. *Computers & Electrical Engineering, 84*, 106606. https://doi.org/10.1016/j.compeleceng.2020.106606

Kirschner, P. A., Sweller, J., Kirschner, F., & Zambrano, R. J. (2018). From cognitive load theory to collaborative cognitive load theory. *International Journal of Computer-Supported Collaborative Learning, 13*, 213–233. https://doi.org/10.1007/s11412-018-9277-y

Kaghat, F. Z., Le Prado, C., Damala, A., & Cubaud, P. (2009). Experimenting with sound immersion in an arts and crafts museum. In *Proceedings of the 8th International Conference Entertainment Computing–ICEC, Paris, France* (pp. 173–178). Springer. https://doi.org/10.1007/978-3-642-04052-8_16

Köster, F., Guse, D., Wältermann, M., & Möller, S. (2015). Comparison between the discrete ACR scale and an extended continuous scale for the quality assessment of transmitted speech. *Fortschritte der Akustik, DAGA, 3*, 150–153.

Krüger, J. M., & Bodemer, D. (2022). Application and investigation of multimedia design principles in augmented reality learning environments. *Information, 13*(2), 74. https://doi.org/10.3390/info13020074

Larsson, P., Väljamäe, A., Västfjäll, D., Tajadura-Jiménez, A., Kleiner, M. (2010). Auditory-Induced Presence in Mixed Reality Environments and Related Technology. In E. Dubois, P., Gray, L., Nigay (Eds.), *The engineering of mixed reality systems* (pp. 143-163). Springer. https://doi.org/10.1007/978-1-84882-733-2_8

Leclère, T., Lavandier, M., & Perrin, F. (2019). On the externalization of sound sources with headphones without reference to a real source. *Journal of the Acoustical Society of America, 146*, 2309–2320. https://doi.org/10.1121/1.5128325

Liu, T. C., Lin, Y. C., Tsai, M. J., & Paas, F. (2012). Split-attention and redundancy effects on mobile learning in physical environments. *Computers & Education, 58*(1), 172–180. https://doi.org/10.1016/j.compedu.2011.08.007

Mayer, R., & Fiorella, L. (2014). Principles for reducing extraneous processing in multimedia learning: Coherence, signaling, redundancy, spatial contiguity, and temporal contiguity principles. In R. Mayer (Ed.), *The Cambridge handbook of multimedia learning* (pp. 279–315). Cambridge University Press.

Mayer, R. E., & Fiorella, L. (2021). *The Cambridge handbook of multimedia learning*. Cambridge University Press. https://doi.org/10.1017/9781108894333

Mayer, R. E., Steinhoff, K., Bower, G., & Mars, R. (1995). A generative theory of textbook design: Using annotated illustrations to foster meaningful learning of science text. *Educational Technology Research and Development, 43*(1), 31–41. https://doi.org/10.1007/BF02300480

McGurk, H., & MacDonald, J. (1976). Hearing lips and seeing voices. *Memory & Cognition, 264*, 746–748. https://doi.org/10.1038/264746a0

Meyerhoff, H. S., & Huff, M. (2016). Semantic congruency but not temporal synchrony enhances long-term memory performance for audio-visual scenes. *Memory & Cognition, 44*, 390–402. https://doi.org/10.3758/s13421-015-0575-6

Miller, J. (1982). Divided attention: Evidence for coactivation with redundant signals. *Cognitive Psychology, 14*(2), 247–279.

Miller, J. F., Lazarus, E. M., Polyn, S. M., & Kahana, M. J. (2013). Spatial clustering during memory search. *Journal of Experimental Psychology: Learning, Memory, and Cognition, 39*(3), 773–781. https://doi.org/10.1037/a0029684

Moreno, R., & Mayer, R. E. (1999). Cognitive principles of multimedia learning: The role of modality and contiguity. *Journal of Educational Psychology, 91*(2), 358–368. https://doi.org/10.1037/0022-0663.91.2.358

Moreno, R., & Mayer, R. E. (2007). Interactive multimodal learning environments. special issue on interactive learning environments: Contemporary issues and trends. *Educational Psychology Review, 19*, 309–326. https://doi.org/10.1007/s10648-007-9047-2

Myrczik, E. P. (2014). Satisfying personal needs at the museum: The role of digital technologies. *MedieKultur: Journal of Media and Communication Research, 57*, 176–196. https://doi.org/10.7146/mediekultur.v30i57.16055

Paas, F., & Sweller, J. (2012). An evolutionary upgrade of cognitive load theory: Using the human motor system and collaboration to support the learning of complex cognitive tasks. *Educational Psychology Review, 24*, 27–45. https://doi.org/10.1007/s10648-011-9179-2

Plass, J. L., & Kaplan, U. (2016). Chapter 7—Emotional design in digital media for learning. In S. Y. Tettegah & M. Gartmeier (Eds.), *Emotions, technology, design, and learning. Emotions and Technology* (pp. 131–161). Academic Press.

Plenge, G. (1972). On the problem of "In Head Localization." *Acta Acustica United with Acustica, 26*(5), 241–252.

Plenge, G. (1974). On the differences between localization and lateralization. *Journal of the Acoustical Society of America, 56*, 944–951. https://doi.org/10.1121/1.1903353

Polyn, S. M., Norman, K. A., & Kahana, M. J. (2009). A context maintenance and retrieval model of organizational processes in free recall. *Psychological Review, 116*(1), 129–156. https://doi.org/10.1037/a0014420

Pouw, W., Rop, G., de Koning, B., & Paas, F. (2019). The cognitive basis for the split-attention effect. *Journal of Experimental Psychology - General, 148*(11), 2058–2075. https://doi.org/10.1037/xge0000578

Schroeder, N. L., & Cenkci, A. T. (2018). Spatial contiguity and spatial split-attention effects in multimedia learning environments: A meta-analysis. *Educational Psychology Review, 30*, 679–701. https://doi.org/10.1007/s10648-018-9435-9

Sommerauer, P., & Müller, O. (2014). Augmented reality in informal learning environments: A field experiment in a mathematics exhibition. *Computers & Education, 79*, 59–68. https://doi.org/10.1016/j.compedu.2014.07.013

Stitt, P., Picinali, L., & Katz, B. F. G. (2019). Auditory accommodation to poorly matched non-individual spectral localization cues through active learning. *Scientific Reports, 9*, 1063. https://doi.org/10.1038/s41598-018-37873-0

Sutojo, S., Thiemann, J., Kohlrausch, A., & van de Par, S. (2020). Auditory Gestalt rules and their application. In J. Blauert & J. Braasch (Eds.), *The technology of binaural understanding Modern acoustics and signal processing*. Springer. https://doi.org/10.1007/978-3-030-00386-9_2

Terrenghi, L., & Zimmermann, A. (2004). Tailored audio augmented environments for museums. In *Proceedings of the 9th international conference on Intelligent user interfaces* (pp. 334–336). https://doi.org/10.1145/964442.964523

Theile, G. (1980). *On the localization in the superimposed soundfield.* [Doctoral Dissertation, Technical University of Berlin]. https://hauptmikrofon.de/theile/1980-2_Diss._Theile_englisch.pdf

Tsepapadakis, M., Gavalas, D., & Koutsabasis, P. (2022). 3D audio + augmented reality + AI chatbots + IoT: An immersive conversational cultural guide. In: L. T. De Paolis, P. Arpaia, M. Sacco (Eds.), *Proceedings of the international conference on extended reality, XR Salento 2022. Lecture Notes in Computer Science, Vol. 13445* (pp. 249–260). Springer. https://doi.org/10.1007/978-3-031-15546-8_22

Yang, J., & Sörös, G. (2019). Audio-augmented museum experiences using wearable visual-inertial odometry. In *Proceedings of the 18th International Conference on Mobile and Ubiquitous Multimedia, Pisa, Italy, 21,* (pp. 1–6). Association for Computing Machinery. https://doi.org/10.1145/3365610.3365616

Yang, X., Wang, F., Mayer, R. E., Hu, X., & Gu, C. (2023). Ocular foundations of the spatial contiguity principle: Designing multimedia materials for parafoveal vision. *Journal of Educational Psychology*. https://doi.org/10.1037/edu0000823

Zhou, Y., Chen, J., & Wang, M. (2022). A meta-analytic review on incorporating virtual and augmented reality in museum learning. *Educational Research Review, 36*, 100454. https://doi.org/10.1016/j.edurev.2022.100454

Zimmermann, A., & Lorenz, A. (2008). LISTEN: A user-adaptive audio-augmented museum guide. *User Modeling and User-Adapted Interaction, 18*(5), 389–416. https://doi.org/10.1007/s11257-008-9049-x

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